

Project Horizon: Drone Swarm Dataset Generation using GMM

Dataset Settings

Number of Records

Generate Dataset

Download CSV

Data

Statistics

Graphs

Dataset Preview

Drone_X	Drone_Y	Victim_X	Victim_Y	Battery	Speed	Distance	Weather	Terrain	Reward	Done
28.76	27.36	27.39	24.17	32.77	8.09	3.47	Rain	Hill	100.00	0.00
78.83	59.39	67.97	76.51	31.56	19.14	20.28	Fog	Forest	-20.28	0.00
40.90	16.02	70.58	31.55	31.93	10.69	33.50	Rain	Forest	-33.50	0.00
88.30	85.34	73.61	75.10	61.15	14.39	17.91	Clear	Urban	-17.91	0.00
94.05	84.77	77.15	83.74	59.43	7.75	16.93	Windy	Forest	-16.93	0.00
4.56	47.79	74.36	86.29	69.31	14.89	79.71	Windy	Plain	-79.71	0.00
52.81	77.37	25.65	26.11	55.79	16.02	58.01	Clear	Hill	-58.01	0.00
89.24	29.54	78.77	24.44	24.45	12.44	11.65	Windy	Forest	-11.65	0.00
55.14	6.56	74.17	28.46	20.43	18.03	29.01	Rain	Plain	-29.01	0.00
45.66	44.05	87.60	14.31	37.75	17.48	51.41	Rain	Urban	-51.41	0.00

Rows : 1000 \nColumns : 11 \nAlgorithm: Gaussian Mixture Model (GMM)

Data

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Summary Statistics

Drone_X		Drone_Y		Victim_X		Victim_Y		Battery		Speed	
Min.	: 0.05	Min.	: 0.12	Min.	: 6.23	Min.	: 5.78	Min.	:20.28	Min.	: 5.000
1st Qu.	:25.36	1st Qu.	:25.32	1st Qu.	:27.98	1st Qu.	:22.75	1st Qu.	:38.68	1st Qu.	: 8.655
Median	:48.99	Median	:49.84	Median	:63.30	Median	:34.53	Median	:59.72	Median	:12.585
Mean	:49.73	Mean	:49.83	Mean	:53.50	Mean	:48.29	Mean	:59.89	Mean	:12.560
3rd Qu.	:74.68	3rd Qu.	:74.55	3rd Qu.	:77.19	3rd Qu.	:74.81	3rd Qu.	:79.67	3rd Qu.	:16.530
Max.	:99.94	Max.	:99.95	Max.	:96.36	Max.	:93.44	Max.	:99.97	Max.	:19.990
Distance		Weather		Terrain		Reward		Done			
Min.	: 2.38	Length:	1000	Length:	1000	Min.	: -114.52	Min.	: 0		
1st Qu.	: 32.05	Class :	character	Class :	character	1st Qu.	: -68.79	1st Qu.	: 0		
Median	: 51.22	Mode :	character	Mode :	character	Median	: -51.22	Median	: 0		
Mean	: 50.79					Mean	: -50.07	Mean	: 0		
3rd Qu.	: 68.79					3rd Qu.	: -32.05	3rd Qu.	: 0		
Max.	:114.52					Max.	: 100.00	Max.	: 0		

Mean Values

Drone_X	Drone_Y	Victim_X	Victim_Y	Battery	Speed	Distance	Reward	Done
49.72762	49.82571	53.49866	48.29128	59.88837	12.55983	50.79063	-50.06506	0.00000

Standard Deviation

Drone_X	Drone_Y	Victim_X	Victim_Y	Battery	Speed	Distance	Reward	Done
28.748445	28.570673	25.555127	27.099183	23.375995	4.410087	23.336492	26.226194	0.000000

Variance

Drone_X	Drone_Y	Victim_X	Victim_Y	Battery	Speed	Distance	Reward	Done
826.47310	816.28338	653.06450	734.36574	546.43716	19.44887	544.59186	687.81323	0.00000

Missing Values

Drone_X	Drone_Y	Victim_X	Victim_Y	Battery	Speed	Distance	Weather	Terrain	Reward	Done
0	0	0	0	0	0	0	0	0	0	0

Correlation Matrix

	Drone_X	Drone_Y	Victim_X	Victim_Y	Battery	Speed	Distance	Reward	Done
Drone_X	1.00	-0.06	0.02	0.01	0.07	-0.02	-0.05	0.04	NA
Drone_Y	-0.06	1.00	-0.06	-0.07	0.05	0.06	0.03	-0.02	NA
Victim_X	0.02	-0.06	1.00	0.29	-0.03	-0.02	0.03	-0.03	NA
Victim_Y	0.01	-0.07	0.29	1.00	0.01	0.02	-0.08	0.08	NA
Battery	0.07	0.05	-0.03	0.01	1.00	-0.06	0.02	-0.04	NA
Speed	-0.02	0.06	-0.02	0.02	-0.06	1.00	0.00	0.00	NA
Distance	-0.05	0.03	0.03	-0.08	0.02	0.00	1.00	-0.95	NA
Reward	0.04	-0.02	-0.03	0.08	-0.04	0.00	-0.95	1.00	NA
Done	NA	NA	NA	NA	NA	NA	NA	NA	1

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Graphs



